

# Operations and Installation Instructions

## 1053 Emulator for the IBM 1130

### 1053 Console Printer on the IBM 1130 System

The IBM 1130 System was announced and delivered in 1965. The system comes with a console that has a keyboard and a printer, along with controls and display lights. The console printer is a print only I/O Selectric typewriter.

The printer uses paper forms with pin feed holes on the two edges, with the typewriter tractor mechanism moving the forms through the machine reliably. Boxes of paper forms in this size are no longer available.

The line width is 120 columns, although the user can adjust the left and right margins to a narrower range. An indicator on the front of the printer shows the current column where the carrier is sitting.

As each character is typed the carrier advances to the right until it reaches the right margin, when a bell will ring and the 1130 will automatically move the carrier back to the left margin and advance the paper one position.

A type element (golfball) is installed on the carrier to provide the 88 characters that can be printed. The normal ball for the IBM 1130 is numbered 969 just under the top latch, although variations were used for some non-US customers.

The 1053 has a dual color ribbon, allowing the selection of either black or red type to be printed on the paper.. This is under software control.

Like any Selectric based typewriter, it provides tab stops to allow the software to move the carrier to predetermined columns. These are set or cleared by manually moving the carrier to the desired column then turning the lever on the front of the printer to set or clear the tab stop.

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The 1053 has a high speed carrier return feature installed which returns from the right side to the left margin in roughly one second. Line feed operations move the paper up one or two lines at a time - selectable by a lever on the printer - in under a second. The typewriter can produce print at a rate of up to 15.6 characters per second.

The front of the printer also has three buttons that allow the user to trigger a carrier return/line feed, tab the carrier to the next tab stop, or space the carrier one column to the right.

The front of the printer also hosts a set of 16 toggle switches, named the Console Entry Switches. These are not related to the console printer at all, other than being mounted on the printer's front panel



image modified from- Martin Skott [File:IBM\\_1130\\_console.ms.jpg](File:IBM_1130_console.ms.jpg) [accessed 2026-06-05]. License/note: CC BY-SA 2.0.

Three cables run from the printer down through the 1130 to plug into connector blocks. Two are for the signals between the 1130 and 1053, the third carries power to the console printer.

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IBM used the same mechanism for the primary operator consoles of their System 360 mainframes, introduced at about the same time. That 1052 had a slightly different case, without the console entry switches of the 1130 and had a keyboard attached.

The keyboard was not a Selectric typewriter part, it was the keyboard used in the 029 Keypunch. It shared the case but was unrelated to the printer portion of the 1052. The same keyboard mechanism is on the IBM 1130 but not physically connected to the console printer.



image: ComputerGeek7066 - [File:IBM 1052.jpg](#) [accessed 2026-06-05]. License/note: CC BY-SA 4.0 (cropped).

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### Overview of the 1053 Emulator

The emulator consists of a box that sits on top of the 1130 system and connected by a USB serial cable to a computer that runs a terminal program where the output of the console printer is displayed and recorded.



The box has a three digit display that shows the column where the carrier is sitting at the moment. Buttons across the bottom allow the user to command the printer to space one column to the right, move the carrier to the next column on the right that has a tab stop set, or to return the carrier to the left and advance the paper one line.

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Buttons on the left will set or clear the tab stop at the current column as displayed on the three digit display. Other interactions are supported by commands through the USB serial cable.

A menu is printed when the emulator starts up, listing the commands that it will recognize when typed on the computer emulator program that displays the output.

The left and right margins are set by commands, otherwise they default to column 1 and column 120. The emulator program displays output in red or black text, based on how the 1053 would be typing at the moment.

For convenience, commands are provided to set or clear tabs by entering the column number, rather than having to space the carrier over to the desired column. Another command will display all the tabs that are set.

One additional type element can be used with the IBM 1130, to run APL (A Programming Language). The ball has 988 on the top near the lever and contains the specialized characters that are used in APL. To switch between typeballs, the command TYPE # is issued, where # is N for the 969 ball and A for the 988 ball.

Some of the characters used in APL are composite, formed by typing one character from the ball, backspacing, and typing over it with another character from the ball. The display will show the composite character just as it would have been produced by ink overtyped on ink using paper in the 1053.

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### Starting the Putty emulator program

The 1053 Emulator has been tested with the free Putty terminal emulator running on Windows. It requires that the Windows system has installed the free APL 385 Unicode truetype font, then configure the terminal to use that font.

Assuming you have configured Putty and saved the configuration, you load that saved config to start the program. The configuration also specifies that a log file is saved with the output. The file does not save colors but otherwise it will match the text shown on the screen.

Generally, the user will set the left and right margins via commands, set any tab stops expected by software and then start the 1130 software.

The 1053 Emulator is powered by the USB connection from the computer running Putty. When it is not on, the FORMS lamp on the 1130 will light, indicating that the 1053 is not able to print because it does not have paper loaded. As soon as the Emulator powers up, the FORMS lamp will turn off.

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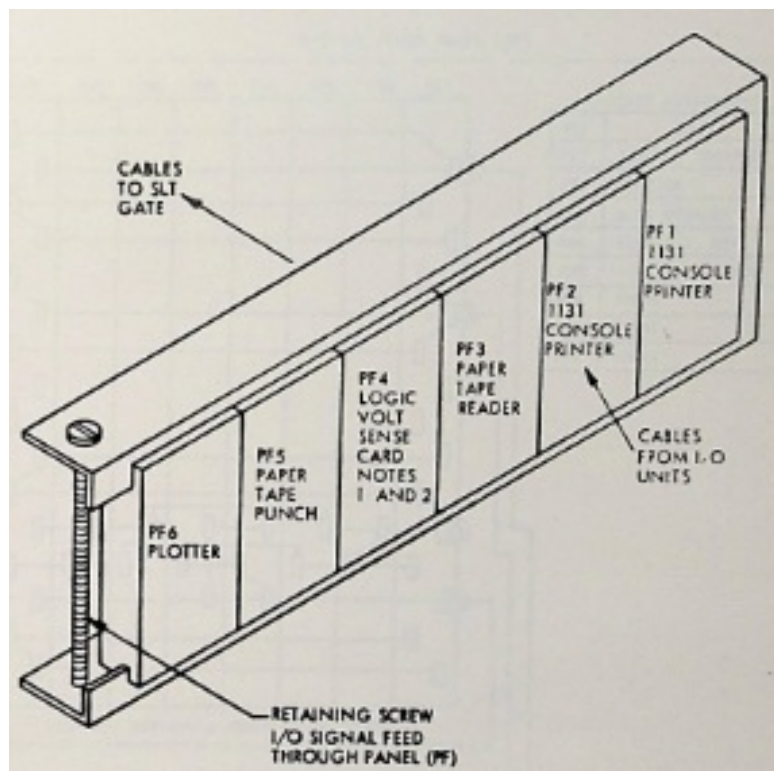
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### Installing the 1053 Emulator signal cables

Inside the 1130, the three cables from the 1053 printer are hooked to connector blocks. The upper connector block is where signals are connected from certain peripherals including the 1053. The lower connector block is where the power cables for the peripherals attach.

The devices that use these connector blocks are older designs that use the 1950s generation Standard Modular System (SMS) rather than the Solid Logic Technology (SLT) of the 360, 1130 and 1960s era devices.

The connector on the cable is an SMS paddle card, which has the same kind of pads on the end as SMS logic cards from machines like the 1401. A female version exists, which is the socket that the logic cards fit into, which can fit on the end of a cable instead of the paddle card. For the 1053, there are two signal cables. PF1 ends in an SMS paddle card (male) while PF2 has an SMS socket (female) on its end.



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The signal connector block has assigned location for each peripheral, with the two rightmost locations used for the 1053. The rightmost position has an adapter installed, so that the PF1 paddle card plugs into one side and a paddle card from the 1130 processor plugs into the rear side. The second position is where the PF2 SMS socket fits, with a paddle card from the 1130 plugged into the socket from the rear.

First, the SMS paddle card from the 1130 logic is pulled out of the PF2 SMS socket from the rear. We unplug the PF1 SMS paddle card from the adapter. The bolt on the left is removed and the PF2 SMS socket is slid out of the bracket to the left to remove the 1053 cables.

The 1053 Emulator PF2 SMS socket is slid into the bracket, the PF1 paddle card is inserted into the adapter in in the rightmost position, and the bolt is replaced in the connector block. The SMS paddle card behind the PF2 socket is inserted from the rear to complete the installation of the signal cables.

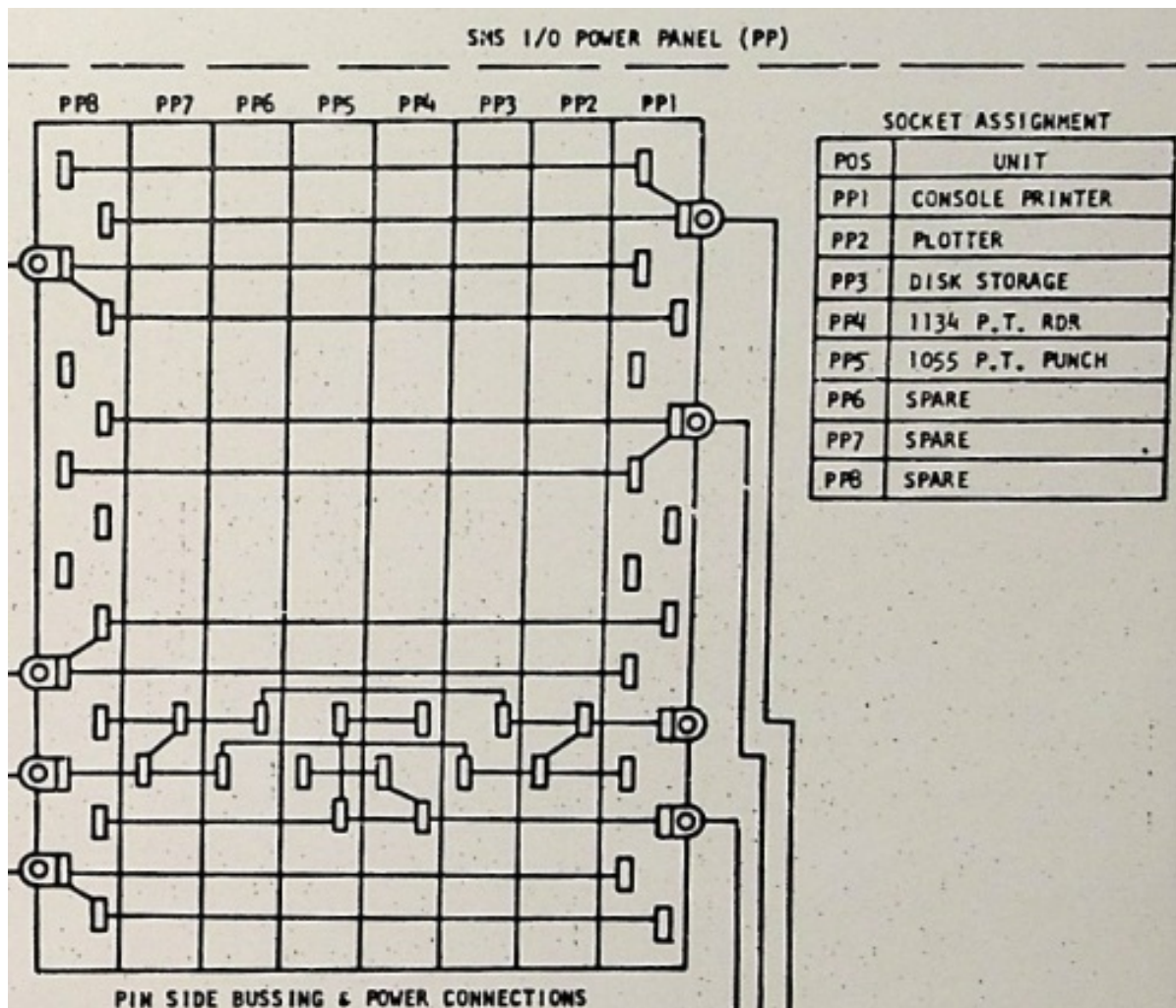
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### Installing the 1053 Emulator power cable

The power connections use a modified version of the SMS paddle card. These have larger combined pads to carry more current, as well as having a notch cut into them that blocks ordinary paddle cards from fitting into the power connector block. The power connector block has a metal bar across the span that prevents signal cards from being inserted.

The power connector block also has assigned locations for each device, with the 1053 assigned to the rightmost socket.

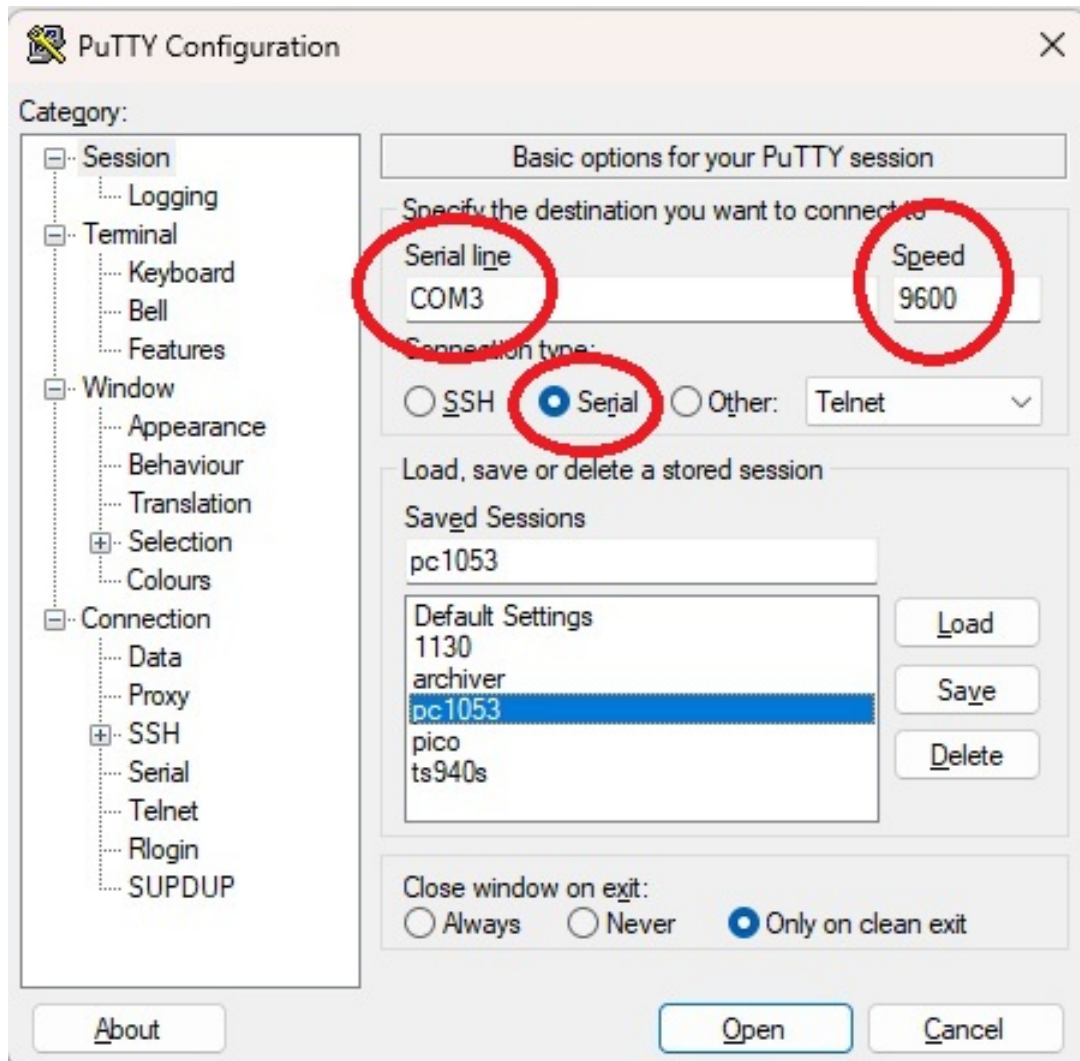


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### Configuration of Putty

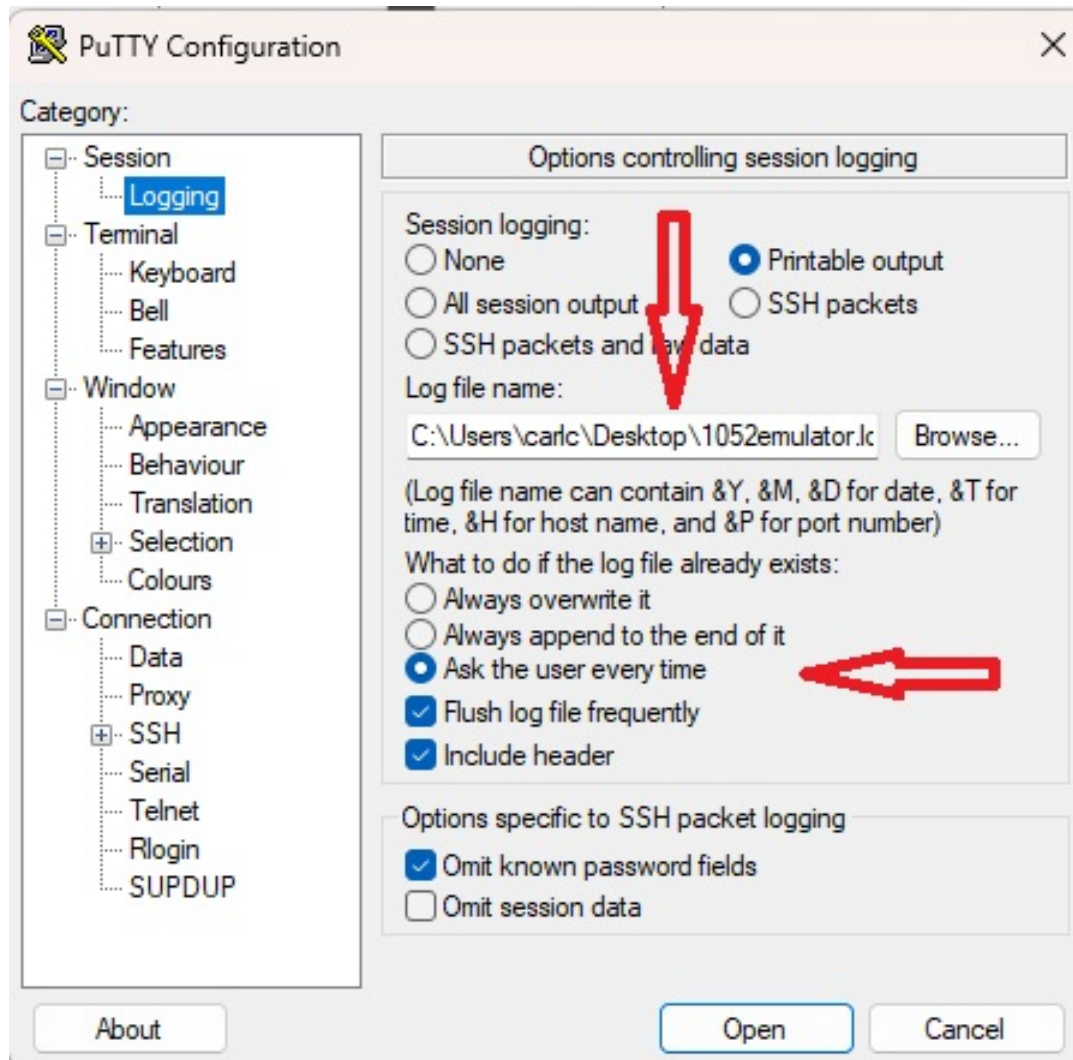
On the main configuration screen, set the connection type to Serial, the speed to 9600 baud and use whatever COM port that the 1053 Emulator shows up with.



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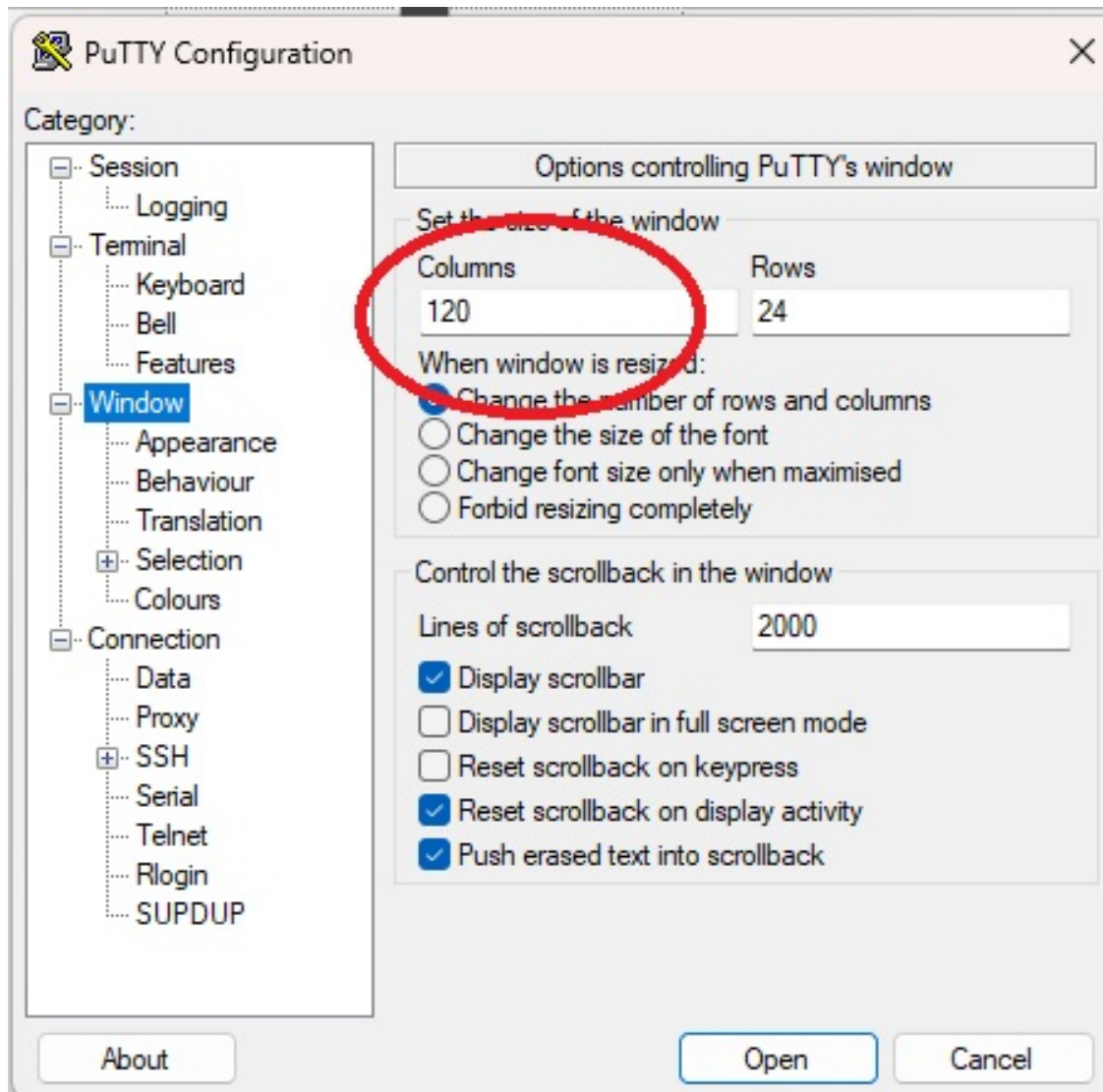
Next, go to the logging screen and choose a file location and name for the log file. Also select the option to ask the user for how to handle it when the log file already exists from a prior run. .



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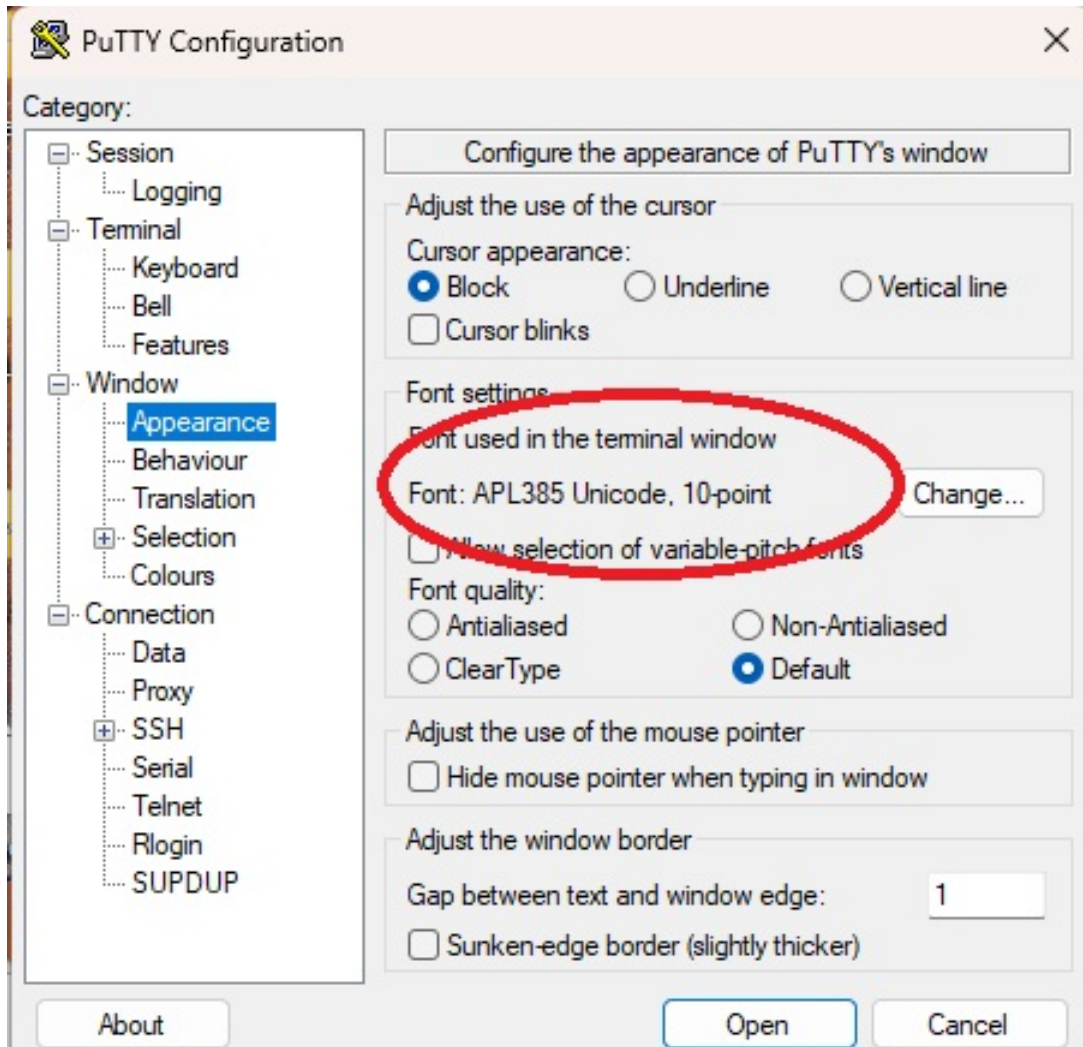
Select the Window category and configure the terminal to 120 column width.



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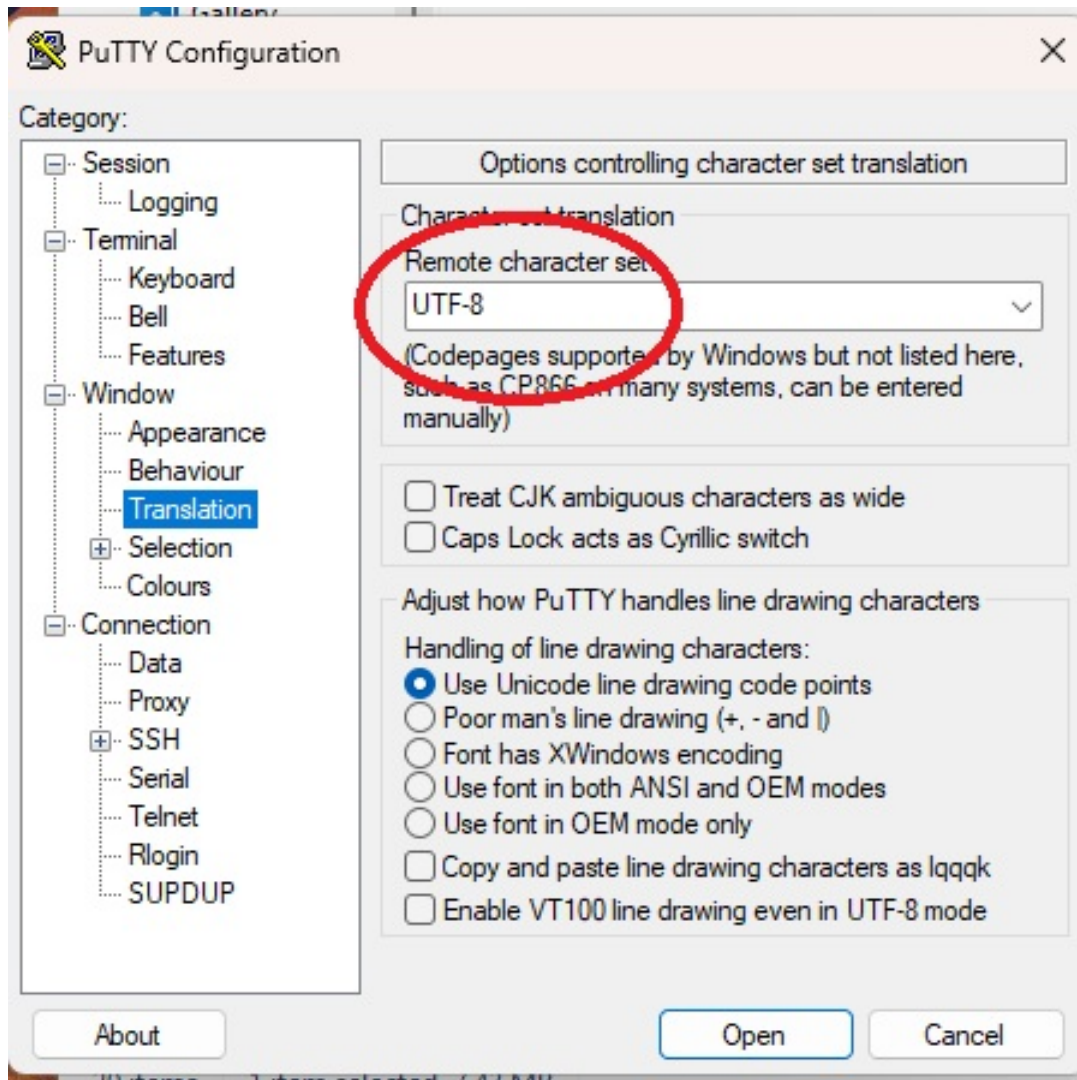
The appearance category is where you change the font for the terminal window to use the APL385 Unicode truetype font, at 10 point size.



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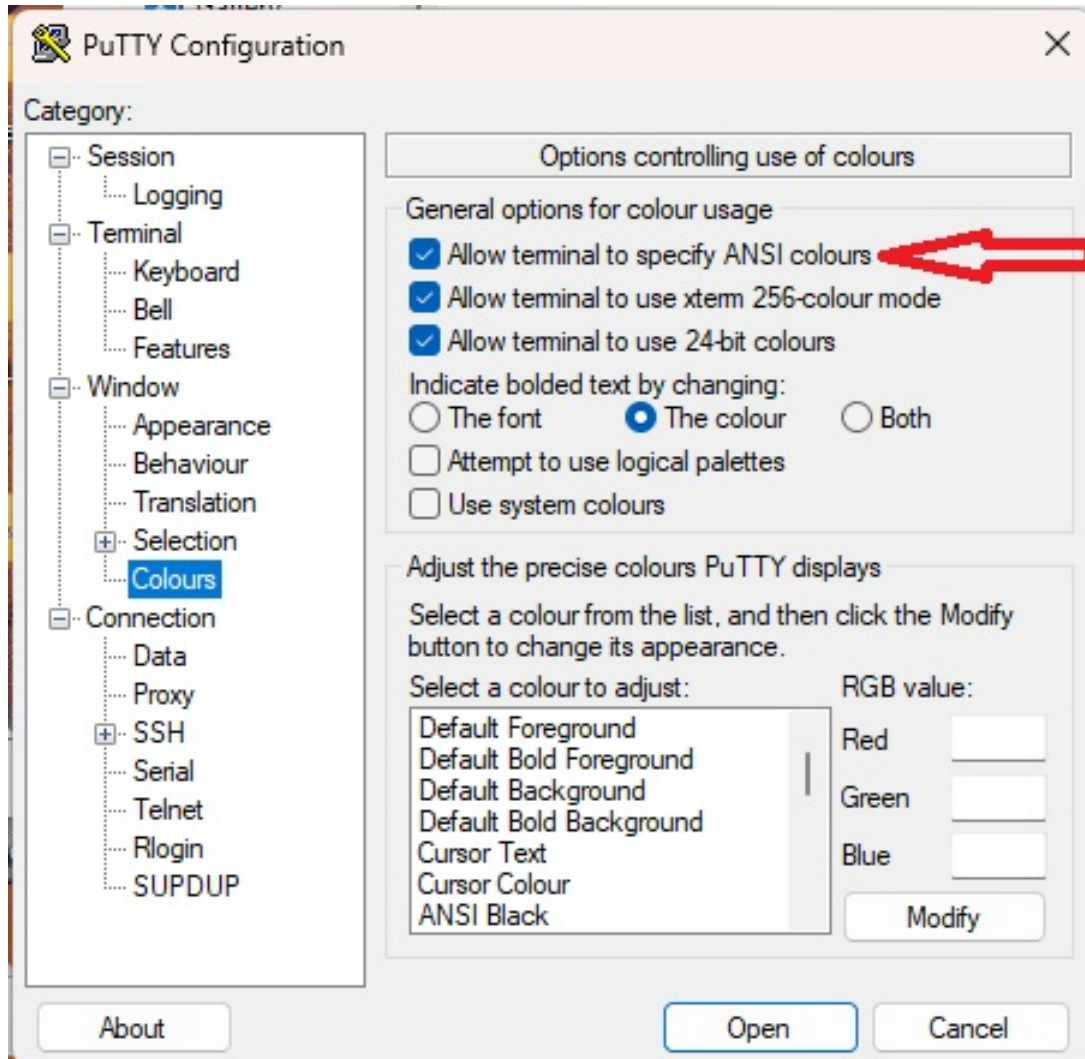
Configure the Translation category to set the character set to UTF-8 which is how the 1053 Emulator is emitting characters.



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The Colors category should set the window to use ANSI Colors which is how the 1053 Emulator controls the text color.



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The final category to look at is Serial. Make sure that the link is set up for 9600 baud, 8 data bits, 1 parity bit, no parity and no flow control.

